

- 5 The Ulin tree (*Eusideroxylon zwageri*), is native to Brunei, Indonesia and the Philippines.

The photograph shows an Ulin tree.



(Source: © Pacific Imagica / Alamy Stock Photo)

- (a) This species of tree is endangered and the export of wood from these trees is forbidden from some countries.

Suggest **three** reasons why this species of tree is endangered.

(3)

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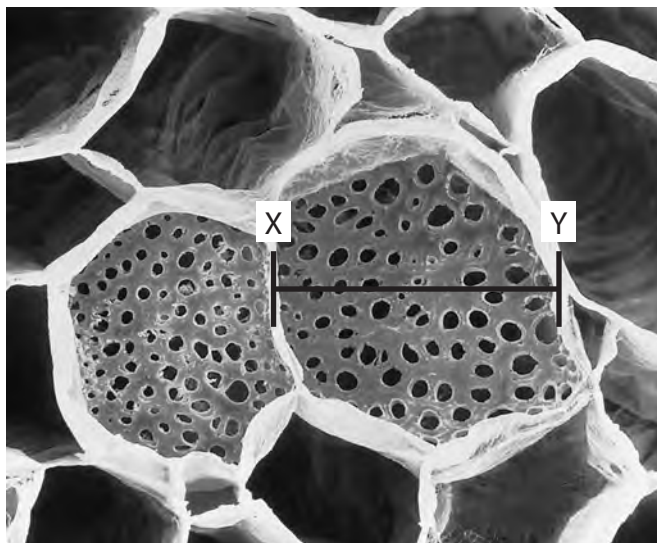
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(b) Trees, and other plants, contain phloem vessels and sclerenchyma fibres.

The photograph shows sieve plates in the phloem of a plant.



(Source: © Science Photo Library / Alamy Stock Photo)

magnification $\times 200$

- (i) Calculate the width of the phloem vessel along the line XY in micrometres.

Give your answer in standard form.

(3)

Answer μm



(ii) Describe the role of phloem.

(2)

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(iii) How many of the following statements about mature sclerenchyma fibres are correct?

- they contain cellulose microfibrils
- they contain pits
- they contain cytoplasm

(1)

☐ **A** none

☐ **B** one

☐ **C** two

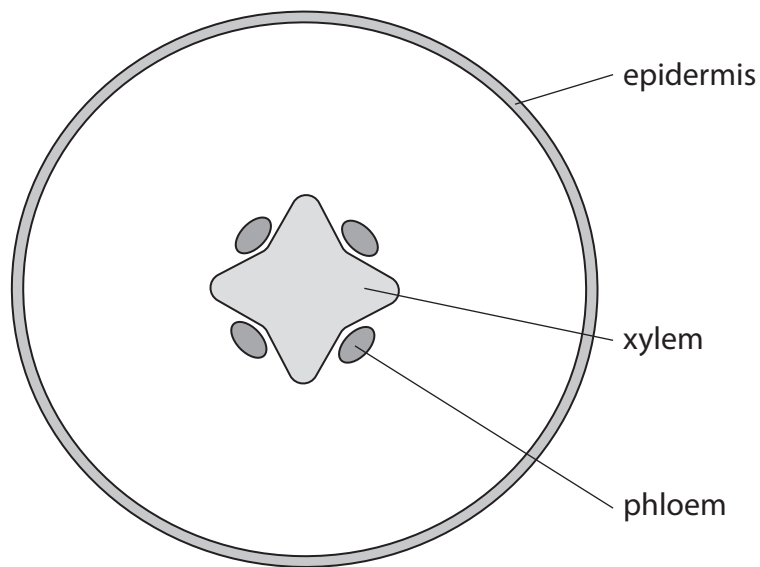
☐ **D** three

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(c) The roots of plants contain phloem and xylem vessels, as shown in the diagram.



Give **three** differences between the distribution of phloem and xylem in the root compared with their distribution in the stem.

(3)

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(Total for Question 5 = 12 marks)

